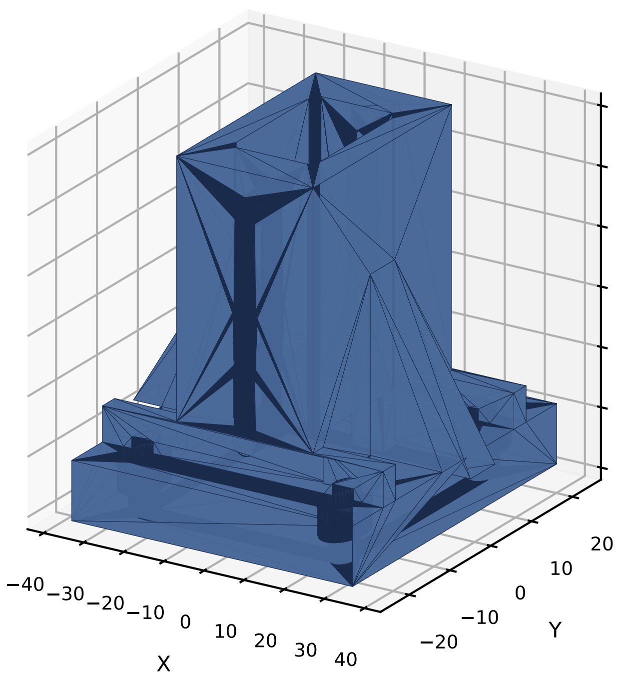
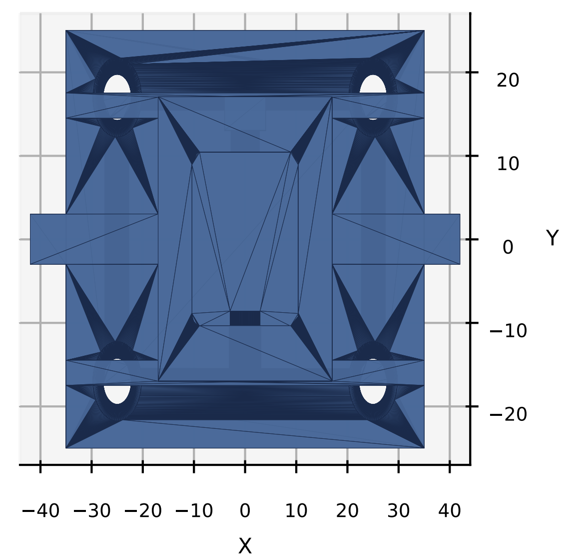
# CSM V3 -- HMI Module 10 -- Mast Base Socket V1.2

*Larger gussets, anti-creep ribs, R1.5 pocket fillets, counterbore for socket-head, 10 mm foot*

### Isometric View




### Top View (X-Y) -- ribs + bolt counterbores

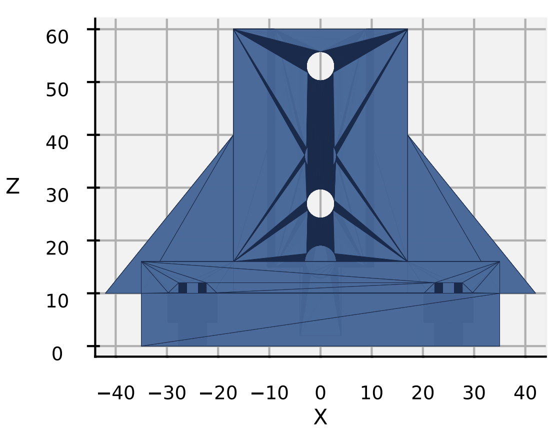



| | |
|---|---|
| **PART** | Mast Base Socket |
| **VERSION** | V1.2 |
| **MODULE** | 10 -- HMI |
| **QTY** | 2 (left + right) |
| **MATERIAL** | PETG (PA12 prod) |
| **INFILL** | 100%, 4 walls |
| **LAYER** | 0.2 mm |
| **MASS** | ~95 g each |
| **TIME** | ~4 h each |
| **INSTALL** | (X=+/-75, Y=-210) |

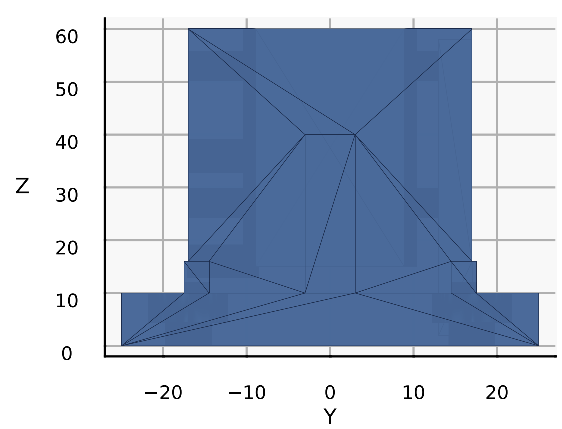
**DIMENSIONS (mm)**

| | |
|---|---|
| Socket outer | 34.0 sq x 50.0 |
| Pocket | 20.8 sq, R1.5 fillets |
| Wall thick | 6.6 |
| Foot | 70.0 x 50.0 x 10.0 |
| Foot bolt CB | 9.5 x 5.5 |
| Gusset | 25.0 x 30.0 x 6 |
| Anti-creep rib | 2x 3.0 x 6.0 |
| Cable channel | 8.0 x 4.0 rear |
| Insert pocket | 6.2 x 8.5 |
| Insert pitch | 4 @ 20.0 |

### Front View (-Y) -- 4x insert pockets @ 20 mm




### Side View (+X) -- larger gusset + rib + cable channel




**V1.2 IMPROVEMENTS**

1. R1.5 pocket inner fillets (printable + easier 2020 insertion + lower stress)
2. Gussets enlarged 18x22 -> 25x30
3. Foot thickness 8 -> 10 mm
4. 2x anti-creep ribs (3 x 6) on foot top
5. Foot bolt CSK -> counterbore (socket-head)
6. Insert pitch 10 -> 20 mm (std patterns)
7. cut_cylinder() helper in macro

ASSEMBLY (per bracket):

1. Print PETG foot-down, 100% infill, 4 walls
2. Heat 4x M5 brass inserts into FRONT pockets
3. 2x M5 T-nuts into 2020 FRONT slot
4. Slide bracket pocket down over 2020 base
5. M5 x 12 socket-heads through FRONT into T-nuts
6. Bracket on wood at (X=+/-75, Y=-210)
7. Drill 4x dia 5.5 through wood under bolt grid
8. M5 x 30 socket-heads through foot CB into wood
9. Washer + M5 nyloc nut under wood; tighten
10. Verify 2020 stands square in both planes